

# Zahra Almadeh

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## SUMMARY

Software Engineering undergraduate at King Fahd University of Petroleum & Minerals (KFUPM) with academic and project-based experience in full-stack software development, database systems, and AI-enabled applications. Proficient in Java, Python, SQL, JavaScript, React.js, Node.js, and MongoDB, with a solid grounding in object-oriented design principles, algorithmic thinking, and systematic problem-solving. Currently seeking a two-month summer training placement to further advance technical competencies and professional skills in the field of software engineering.

## EDUCATION

### B.S. in Software Engineering

King Fahd University of Petroleum & Minerals (KFUPM)

Aug 2022 – Jun 2027 (Expected)

Dhahran, SA

## TECHNICAL SKILLS

- **Programming Languages:** Java, Python, SQL, JavaScript, HTML, CSS
- **Web Development:** React.js, Node.js, MongoDB, REST APIs, Front-End Development, Full-Stack Development
- **Software Engineering:** Object-Oriented Programming (OOP), Design Patterns, UML, System Design, Agile Development, SDLC
- **Data & AI:** Machine Learning, Deep Learning, Neural Networks, Data Preprocessing, Model Evaluation
- **Databases:** Relational Databases, SQL, Database Design, Database Normalization (BCNF)
- **Tools & Technologies:** Git/GitHub, JavaFX, SQL Workbench, Excel

## PROJECTS

### Makhzan Website

- Developed a full-stack inventory management system for medical facilities using React.js, Node.js, and MongoDB
- Built responsive front-end interfaces with search, filtering, and inventory tracking functionalities
- Implemented secure back-end services for authentication, inventory management, and automated low-stock notifications

### Airplane Accident Severity Classification

- Built a Deep Neural Network (MLP) trained on 10,000+ records, achieving 92.3% classification accuracy
- Performed data preprocessing, feature scaling, and 5-fold cross-validation for model optimization
- Benchmarked model performance against KNN and SVM algorithms using Python-based machine learning workflows

### Course Scheduling & Route Planner

- Developed a Java-based scheduling and route planning system supporting multi-course scheduling across locations
- Implemented Composite and Strategy design patterns using object-oriented programming principles
- Designed scalable system architecture and UML models for maintainable software development

### Horse Campus Events Database System

- Designed a BCNF-normalized relational database system covering 10+ entities and relationships
- Developed SQL queries, procedures, triggers, and views for efficient data management and reporting
- Validated database functionality through structured testing and query execution

## EXTRACURRICULAR ACTIVITIES

### Google Developer Student Club (KFUPM)

- Developed branding and visual content for technical events

Jan 2024 – May 2024

### Female Computer Science Club

- Assisted in marketing and promotional campaigns

Nov 2024 – May 2025

### TEDxKFUPM

- Designed visual materials for a large-scale university event

Sep 2025 – Feb 2026

### VentureCraft

- Selected through peer nomination to join the design team

March 2026 – present